



DIFFERENTIAL EQUATIONS

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Course contents

Formation of differential equations, solution of various types of first order differential equations, orthogonal trajectories, application in physical problems. Linear differential equations of second order, complementary function and particular integral. Solution of non-homogeneous linear differential equations of second order and higher by:

- (i) the method of undetermined coefficients;
- (ii) the method of variation of parameters;
- (iii) the method of power series.

Application of second order differential equations, system of differential equations.

Laplace transform, solution of initial value problems by Laplace transform. Convolution theorem and applications.

Formation of partial differential equations, equations reducible to ordinary differential equations, equations of the form $P_p + Q_q = R$, solution by the method of separation of variables. Wave, Heat, and Laplace equations.



Course contents

Periodic functions. Even and odd functions. Fourier series of functions of period 2 and arbitrary period. Half range series. Partial differential equations and Fourier series. Sturm-Liouville problems.

Pre-requisites: This course is intended for those students who have studied single-variable calculus, i.e., single-variable calculus is pre-requisite. Whereas, multi-variable calculus is a co-requisite, i.e., students could take Differential equations and multi-variable calculus simultaneously.



Recommended books

The following are considered as good reference books:

- 1 Erwin Kreyszig (latest edition), "Advanced engineering mathematics".
- 2 D. G. Zill and M. R. Cullen (latest edition), "Differential equations with boundary value problems".
- 3 D. G. Zill (latest edition), "A first course on differential equations with modelling applications".
- 4 E. A. Bender (latest edition), "An introduction to mathematical modelling".
- 5 N. A. Shah "Ordinary differential equations for scientists and engineers".
- 6 N. A. Shah "Partial differential equations for scientists and engineers".

Acknowledgement. Much of the material below is from different resources, in-particular, from the above recommended books.



Learning outcomes

This course is an introduction to **differential equations**. The primary objective of this course is to provide students a modern introduction to identify different types of differential equations and select and apply the appropriate analytical technique for finding the solution of first order and selected higher order differential equations. It is vital to realize that learning how to apply mathematics is different from learning mathematics. There is less emphasis on building-up theory, and more emphasis on building-up the skills needed to formulate and solve differential equations.

By the end of this course, the key things students will learn are:

- 1 Solve differential equations of first and higher orders using different techniques;
- 2 Use Laplace transforms to solve initial value problems;
- 3 Develop the understanding of separation of variables to solve PDEs;
- 4 Use Fourier transforms to solve boundary value problems.



Expectations and workload

Differential equations is assumed to use at least 10 hours per week of students time. The expected normal pattern of student study each week is

- 1 three hours of lectures;
- 2 three hours of lecture preparation and revision (reading the text books);
- 3 four hours of assignments and/or home work and preparing for test and exam.

Students are expected to attend all lectures and to come prepared. This means that students will have previewed the material and done any preliminary examples that have been set. Stay on top of the course material as it is covered. If you get behind, it could be difficult to catch up. Make use of the assistance available; ask for help as soon as you need it. Students are encouraged to collaborate with one another and to work together.

P.S. If you are in any doubt about the permissible degree of collaboration, please do not hesitate to come and see me in person.



Some basic concepts of differential equations

Differential equations (DEs) occur in the mathematical formulation of many physical phenomena in science and engineering. For example:

- 1 Determining the motion of projectile, rocket, satellite or planet.
- 2 Finding the charge or current in an electric circuit.
- 3 Study of chemical reactions.
- 4 Determination of curves with given geometrical properties.

Differential equations are of fundamental importance in science and engineering, because many physical laws and relations appear mathematically in the form of such equations. In other words, differential equations are the language in which the laws of nature are expressed. Understanding properties of solutions of differential equations is fundamental to much of contemporary science and engineering. Ordinary differential equations (ODEs) deal with functions of one variable, which can often be thought of as time.



Some basic concepts of differential equations

Basically, a differential equation is a relation between a function and its derivatives. Differential equations form the language in which the basic laws of science are expressed. The science tells us how the system at hand changes “from one instant to the next”. The challenge addressed by the theory of differential equations is to take this short-term information and obtain information about long-term overall behavior. So the art and practice of differential equations involves the following sequence of steps:

- 1 First of all, we “model” a system (physical, chemical, biological, economic, or even mathematical) by means of a differential equation;
- 2 then we attempt to gain information about solutions of this equation;
- 3 and finally, we translate this mathematical information back into the scientific context.



Some basic concepts of differential equations

Figure:- A schematic.

To give a basic example, one of the most important laws in mechanics is Newton's second law of motion which states that the rate of change of momentum of a body is proportional to the force applied to it. For motion in a straight line, the mathematical statement of this law is

$$F = ma,$$

where, a = acceleration is the second derivative of x = position. Forces don't effect x directly, but only through its derivatives. This is a second order ODE, and we will study second order ODEs extensively later in the course.



Some basic concepts of differential equations

Definition.

Ordinary differential equations (ODEs).

Partial differential equations (PDEs).

Order of a differential equation.

Degree of a differential equation.

Linear and non-linear differential equations.

A solution (integral) of a differential equation.

Initial and boundary conditions.

Initial and boundary value problems.



Some basic concepts of differential equations

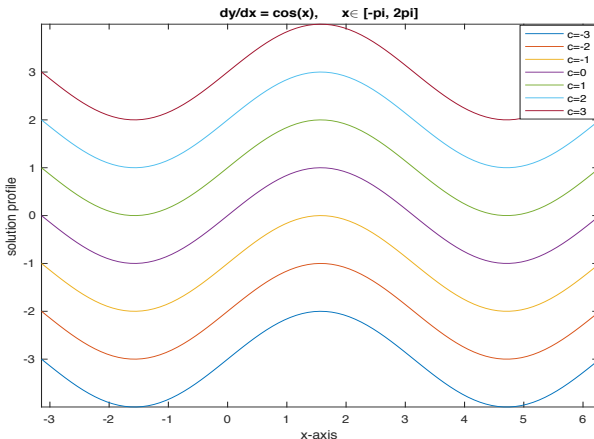


Figure: Solutions $y = \sin(x) + c$ of the ODE $y' = \cos(x)$.



Some basic concepts of differential equations

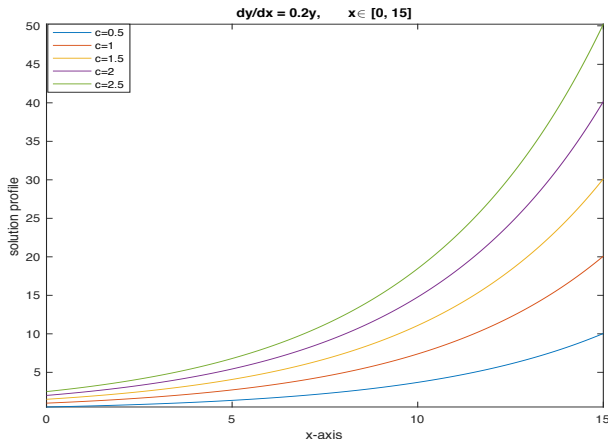


Figure: Solutions of the ODE $y' = 0.2y$ (exponential growth).



Some basic concepts of differential equations

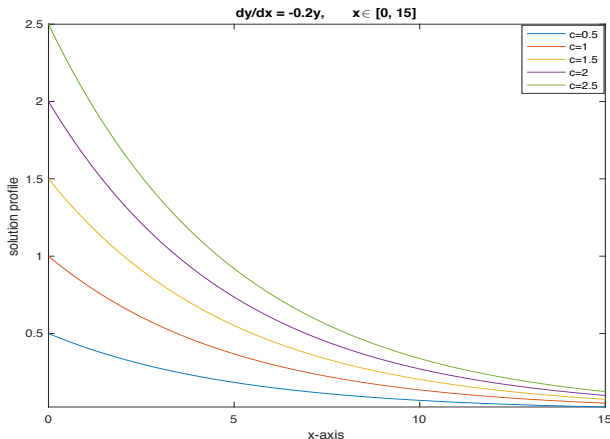


Figure: Solutions of the ODE $y' = -0.2y$ (exponential decay).



Some basic concepts of differential equations

Direction fields

Basically, direction field (slope field) serves as a useful mathematical tool in understanding the behavior of solutions of a differential equation. In other words, direction field, is a way of graphically representing the solutions of a first-order differential equation without actually solving the equation.

The direction field is defined as the collection of small line segments passing through various points having a slope that will satisfy the given differential equation at that point. The actual family of curves (solutions of the differential equation) must have a direction at each point that agrees with that of the line segment of the direction field at that point, so that this method is valuable for gaining some feeling for the behaviour of the solutions in cases in which the equation is difficult to solve or in which the solution is a complicated function.



Some basic concepts of differential equations

Definition:-

A direction field, also called a slope field, is a graphical aid for understanding a differential equation, formed by:

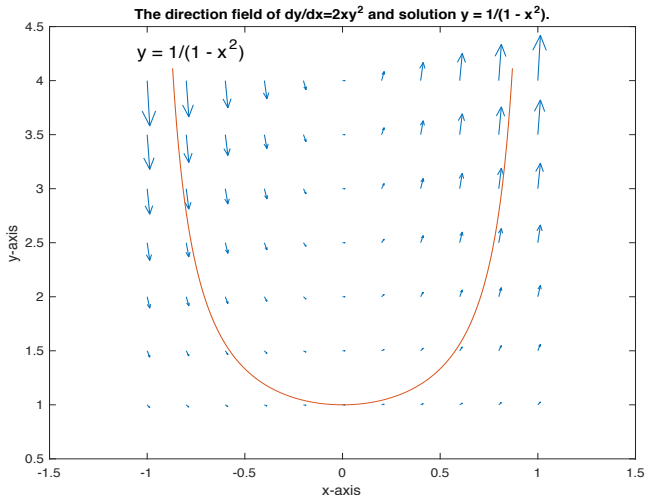
- 1 choosing a grid of points;
- 2 at each point, computing the slope given by the differential equation;
- 3 at each point, drawing a short line segment with that slope.

Remarks:-

- 1 The differential equation $y' = f(x, y)$ gives a direction associated with each point (x, y) in the plane that must be satisfied by any solution curve passing through that point.
- 2 At each point in a direction field, a line segment appears whose slope is equal to the slope of a solution to the differential equation passing through that point.

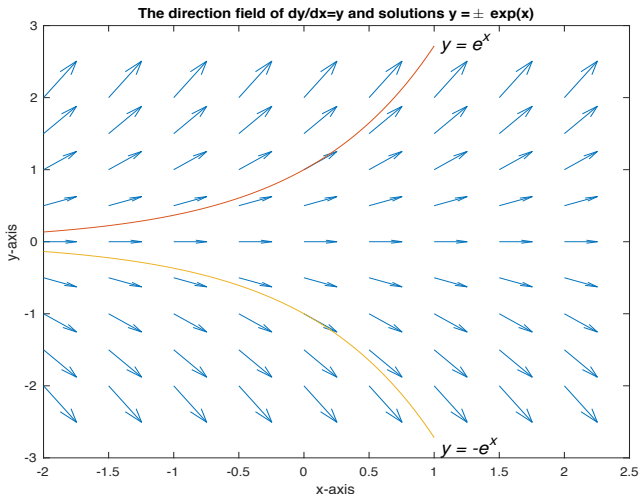


Some basic concepts of differential equations





Some basic concepts of differential equations





Some basic concepts of differential equations

Formation of differential equations

Let x and y are the independent and dependent variables, respectively. Then the equation

$$f(x, y, c) = 0,$$

containing one arbitrary constant c represents a **family of curves**. For example

$$x^2 + y^2 = a^2,$$

where a is arbitrary constant, represents a family of circles with centre at origin and radius a . Similarly

$$f(x, y, c, d) = 0,$$

containing two arbitrary constants c and d also represents a family of curves.



Some basic concepts of differential equations

To form a differential equation from one parameter family of curves

$$f(x, y, c) = 0,$$

we need two equations. One equation is the given equation and the second equation is obtained by differentiating the given equation w.r.t. x . On eliminating the arbitrary constant c from the two equations, we obtain an equation containing x, y , and y' which is a first order differential equation.

Generally, if an equation contains n arbitrary constants, differentiate it successively n times to get n additional equations containing n arbitrary constants and derivatives. Now eliminate n arbitrary constants from the $(n + 1)$ equations to get a differential equation of order n .

Example:- Find the differential equation of

$$y = A \sin x + B \cos x.$$



Some basic concepts of differential equations

Example:- Find the differential equation of family of circles having radius 3 and touching x-axis line.

Example:- Find the differential equation of family of circles having radius 3 and touching y-axis line.

Example:- Find the differential equation of family of circles having centre at x-axis and radius 3.

Example:- Find the differential equation of family of circles having centre at y-axis and radius 3.

Example:- Find the differential equation of family of all circles passing through origin.



Types of ordinary differential equations

Separable ODEs

The differential equation of the form

$$y' = \frac{F(x)}{G(y)}, \quad (1)$$

is called as variable separable ODE, because it can be reduced to the form

$$G(y)y' = F(x) \quad (2)$$

by purely algebraic manipulations. Integrating both sides the above equation with respect to x , we have

$$\int G(y)y' dx = \int F(x) dx + c \quad (3)$$

$$\int G(y) dy = \int F(x) dx + c, \quad \because y' dx = dy \text{ (by calculus)} \quad (4)$$



Types of ordinary differential equations

If $F(x)$ and $G(y)$ are continuous functions then the integrals in (4) exist and by evaluating them we obtain a general solution of (1). This method of solving ODEs is called the method of **separating variables**, and the equation (1) is called as separable equation, because in (4) the variables are now separated, i.e., x appears only on the right and y only on the left.

Example:-

Solve the following differential equation

$$\frac{dy}{dx} = \frac{x^2}{y(1+x^3)}.$$

Exercise:-

Exercise #1.3, Page # 18 (Ref-book no. 1).



Types of ordinary differential equations

Exact ODEs

Recall from calculus that if a function $u(x, y)$ has continuous partial derivatives, then its differential (or total differential) is given by

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy.$$

Note:- if $u(x, y) = c = \text{constant}$, then $du = 0$.

For example, if $u(x, y) = x + x^2 y^3 = c$, then we have

$$du = (1 + 2xy^3)dx + 3x^2 y^2 dy = 0$$

or

$$y' = \frac{dy}{dx} = -\frac{1 + 2xy^3}{3x^2 y^2}.$$

The above equation is an ordinary differential equation that can be solved by going backward.



Types of ordinary differential equations

This idea leads us to a powerful solution method described as follows.
A first-order ODE $M(x, y) + N(x, y)y' = 0$, also written as

$$M(x, y)dx + N(x, y)dy = 0$$

is called an **exact differential equation** if the differential form $M(x, y)dx + N(x, y)dy$ is exact, that is, this form is the differential

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy$$

of some function $u(x, y)$. Then the exact differential equation $M(x, y)dx + N(x, y)dy = 0$ can be written

$$du = 0.$$

By integration, we immediately obtain the general solution of exact differential equation in the form

$$u(x, y) = c.$$



Types of ordinary differential equations

Note:- By comparing the total differential of $u(x, y)$ and $M(x, y)dx + N(x, y)dy = 0$, we see that for an exact differential equation, we have

$$\frac{\partial u}{\partial x} = M,$$

$$\frac{\partial u}{\partial y} = N.$$

From this, we can derive a formula for checking whether $M(x, y)dx + N(x, y)dy = 0$ is exact or not, as follows.

Let M and N be continuous and have continuous first partial derivatives in a region in the xy -plane whose boundary is a closed curve without self-intersections. Then by partial differentiation of the above equations, we have

$$\frac{\partial M}{\partial y} = \frac{\partial^2 u}{\partial y \partial x}, \quad \frac{\partial N}{\partial x} = \frac{\partial^2 u}{\partial x \partial y}.$$



Types of ordinary differential equations

By the assumption of continuity the two second partial derivatives are equal, i.e., $\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y}$, we have

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

This condition is not only necessary but also sufficient for $M(x, y)dx + N(x, y)dy = 0$ to be an exact differential equation.

Note:- To find the general solution of an exact differential equation, we can also use the following formula

$$\int M(x, y) dx + \int [\text{terms of } N(x, y) \text{ free from } x] dy = c.$$

Example:- Check the exactness and then solve the following differential equation

$$(y + 2xy^2) dx + (x + 2x^2y) dy = 0.$$



Types of ordinary differential equations

Non-exact ODEs (Equations reducible to exact form)- Integrating factors

Sometimes a given differential equation is not exact, but can be made exact by multiplying it by some suitable factor, called the **integrating factor (I. F.)**. For example, consider the following differential equation

$$-y \, dx + x \, dy = 0.$$

Here, $M = -y$ and $N = x$, so that

$$\frac{\partial M}{\partial y} = -1,$$
$$\frac{\partial N}{\partial x} = 1.$$

Which shows that the given differential equation is not exact.



Types of ordinary differential equations

However, if we multiply it by $\frac{1}{x^2}$, we get

$$\frac{-y}{x^2} dx + \frac{x}{x^2} dy = 0.$$

Now, $M = \frac{-y}{x^2}$ and $N = \frac{x}{x^2}$. Therefore, we have

$$\frac{\partial M}{\partial y} = -\frac{1}{x^2} = \frac{\partial N}{\partial x}.$$

Hence the given differential equation has become exact. Write this exact equation as

$$d\left(\frac{y}{x}\right) = 0.$$

Integrating the above equation, we get $\frac{y}{x} = c$ as the general solution of the given differential equation. Thus we see that an integrating factor is a factor which changes a non-exact differential equation into an exact differential equation.



Types of ordinary differential equations

Rules for finding integrating factors

- 1 If $M(x, y) dx + N(x, y) dy = 0$ in non-exact differential equation and

$$\frac{M_y - N_x}{N} = F(x),$$

then integrating factor is

$$I. F = e^{\int F(x) dx}$$

- 2 If $M(x, y) dx + N(x, y) dy = 0$ in non-exact differential equation and

$$\frac{N_x - M_y}{M} = F(y),$$

then integrating factor is

$$I. F = e^{\int F(y) dy}$$



Types of ordinary differential equations

- 3 If $M(x, y) dx + N(x, y) dy = 0$ is homogeneous and $xM + yN \neq 0$, then integrating factor is

$$I. F = \frac{1}{xM + yN}.$$

- 4 If $M(x, y) dx + N(x, y) dy = 0$ is non-exact and can be written in the form

$$yf(xy) dx + xg(xy) dy = 0,$$

and if $xM - yN \neq 0$, then integrating factor is

$$I. F = \frac{1}{xM - yN}.$$

Exercise:-

Exercise #1.4, Page # 26 (Ref-book no. 1).



Types of ordinary differential equations

Linear ODEs

Linear ODEs or ODEs that can be transformed into linear ODEs are models of various phenomena; for instance, in physics, biology, population dynamics, and ecology. A first-order ODE is said to be **linear** in the dependent variable y if it can be written in the form

$$\frac{dy}{dx} + P y = Q,$$

where P and Q are functions of x (or constants). If $Q = 0$ for all x in the interval in which we consider the equation, then the above equation is said to be **homogeneous** otherwise, it is said to be **non-homogeneous** or **inhomogeneous** linear ODE. The integrating factor (I. F) of a non-homogeneous linear ODE is

$$\text{I. F} = e^{\int P(x) dx}.$$



Types of ordinary differential equations

Example:- Solve the following differential equation

$$\frac{dy}{dx} + \frac{3y}{x} = 6x^2.$$

Non-linear ODEs (Equations reducible to linear form)- Bernoulli equation

Certain non-linear ordinary differential equations can be reduced to linear form. Practically, the most famous of these equations is the Bernoulli's equation which has the form

$$\frac{dy}{dx} + P y = Q y^n, \quad n \neq 0, 1,$$

where P and Q are functions of x (or constants). In order to reduce this equation into linear differential equation, divide it by y^n , we have

$$y^{-n} \frac{dy}{dx} + P y^{1-n} = Q.$$



Types of ordinary differential equations

Multiplying the above equation both sides by $(1 - n)$, we get

$$(1 - n)y^{-n} \frac{dy}{dx} + (1 - n)P y^{1-n} = (1 - n)Q.$$

Now, put $y^{1-n} = v$ and $(1 - n)y^{-n} \frac{dy}{dx} = \frac{dv}{dx}$, we have

$$\frac{dv}{dx} + (1 - n)P v = (1 - n)Q,$$

the corresponding linear differential equation in v .

Example:- Solve the following differential equation

$$\frac{dy}{dx} + y = xy^3.$$

Exercise:-

Exercise #1.5, Page # 34 (Ref-book no. 1).



Types of ordinary differential equations

Orthogonal trajectories

An important type of problem in physics or geometry is to find a family of curves that intersects a given family of curves at right angles. The new curves are called orthogonal trajectories of the given curves (and conversely).

Let $f(x, y, c) = 0$ and $g(x, y, k)$ are two families of curves with parameters c and k , respectively. If each curve in either family is intersected orthogonally by every curve in the other family, then each family is said to be orthogonal trajectory of the other. Recall from calculus that two curves are said to be orthogonal if their tangents at the point of intersection are perpendicular to each other. For example, the families of the following curves

$$\begin{aligned}x^2 + y^2 = c^2 &\implies f(x, y, c) \equiv x^2 + y^2 - c^2 = 0, \text{ and} \\y = kx &\implies g(x, y, k) \equiv y - kx = 0,\end{aligned}$$

are orthogonal as illustrated in the following figure.



Types of ordinary differential equations

Example:-

Find the orthogonal trajectories for

$$x^2 - y^2 = c.$$

Exercise:-

Exercise #1.6, Page # 38 (Ref-book no. 1).